Analysis classes & CRC cards

Object Oriented Analysis — Identification of Analysis Classes (Noun Convention Method) & CRC cards

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Analysis classes are identified from the use case specifications using the Noun Convention method: nouns are listed, redundant nouns are removed, vague nouns irrelevant to the domain are removed, and any noun representing a characteristic or property of a class is discarded as a class and included instead as an attribute of that class. Classes are then refined progressively through Domain Analysis as each subsequent use case is considered.

DR-001 Manage Scenario Data

Analysis classes identified (Noun Convention) : Scenario, Area, ResourceBase

Refinement of classes through Domain Analysis:

Scenario (ScenarioId, Name, Region, CreatedAt)

Area (AreaId, ScenarioId, Name, Location, Population, Severity, Demand)

ResourceBase (BaseId, ScenarioId, Name, Location)

Resource (ResourceId, BaseId, ResourceType, QuantityAvailable)

DR-002 View Map

Analysis classes identified (Noun Convention): MapView

Refinement of classes through Domain Analysis:

MapView (ScenarioId)

DR-003 Run Baseline Allocation

Analysis classes identified (Noun Convention): TravelTimeMatrix, BaselineAllocator, AllocationRun, Allocation, Metrics

Refinement of classes through Domain Analysis:

TravelTimeMatrix (ScenarioId, BaseId, AreaId, TravelTime)

BaselineAllocator (MethodName)

AllocationRun (RunId, ScenarioId, Method, CreatedAt)

Allocation (AllocationId, RunId, BaseId, AreaId, QuantitySent)

Metrics (RunId, AverageResponseTime, MaxResponseTime, CoverageRatio, UnmetDemand, Utilization, TotalTravelCost)

DR-004 Run Optimized Allocation

Analysis classes identified (Noun Convention): OptimizedAllocator

Refinement of classes through Domain Analysis:

OptimizedAllocator (TravelTimeWeight, UnmetDemandWeight)

AllocationRun (RunId, ScenarioId, Method, CreatedAt, SolverRuntime)

DR-005 Compare Results

Analysis classes identified (Noun Convention): Comparison

Refinement of classes through Domain Analysis:

Comparison (ScenarioId, BaselineRunId, OptimizedRunId, DifferencePerMetric, BetterWhen)

DR-006 Calculate Travel Time

Analysis classes identified (Noun Convention): (No new analysis classes — TravelTimeMatrix attributes extended)

Refinement of classes through Domain Analysis:

TravelTimeMatrix (ScenarioId, BaseId, AreaId, TravelTime, Distance, IsFallback)

Consolidated Analysis Classes (DRRAS)

Scenario (ScenarioId, Name, Region, CreatedAt)

Area (AreaId, ScenarioId, Name, Location, Population, Severity, Demand)

ResourceBase (BaseId, ScenarioId, Name, Location)

Resource (ResourceId, BaseId, ResourceType, QuantityAvailable)

MapView (ScenarioId)

TravelTimeMatrix (ScenarioId, BaseId, AreaId, TravelTime, Distance, IsFallback)

BaselineAllocator (MethodName)

OptimizedAllocator (TravelTimeWeight, UnmetDemandWeight)

AllocationRun (RunId, ScenarioId, Method, CreatedAt, SolverRuntime)

Allocation (AllocationId, RunId, BaseId, AreaId, QuantitySent)

Metrics (RunId, AverageResponseTime, MaxResponseTime, CoverageRatio, UnmetDemand, Utilization, TotalTravelCost)

Comparison (ScenarioId, BaselineRunId, OptimizedRunId, DifferencePerMetric, BetterWhen)

CRC cards :

Scenario	
Responsibility	**Collaboration**
• Hold a disaster scenario with its name and region • Add or load its areas and resource bases • Keep track of the runs made on it	• Area • ResourceBase • MapView • TravelTimeMatrix • AllocationRun • Comparison

Area	
Responsibility	**Collaboration**
• Store location, population, severity and demand • Validate severity and non-negative values • Report severity and demand to the allocators and the map	• Scenario • MapView • BaselineAllocator • OptimizedAllocator • TravelTimeMatrix

ResourceBase	
Responsibility	**Collaboration**
• Store its name and location • Hold resources and report remaining stock • Reduce stock when resources are sent out	• Scenario • Resource • MapView • TravelTimeMatrix • BaselineAllocator • Allocation

Resource	
Responsibility	**Collaboration**
• Store resource type and quantity available • Reject a negative quantity • Report the quantity available	• ResourceBase • OptimizedAllocator

MapView	
Responsibility	**Collaboration**
• Show each area coloured by severity • Show each base as a marker • Display the scenario before any run	• Scenario • Area • ResourceBase

TravelTimeMatrix	
Responsibility	**Collaboration**
• Request travel time for every base-area pair • Fall back to straight-line distance and flag the entry • Store once per scenario and reuse • Return travel time for one pair	• OSRM (actor) • ResourceBase • Area • Scenario • BaselineAllocator • OptimizedAllocator • Metrics

BaselineAllocator	
Responsibility	**Collaboration**
• Sort areas by severity, most severe first • Find the nearest base that still has stock • Assign resources up to the area's demand • Create the run and ask for its metrics	• Area • ResourceBase • TravelTimeMatrix • Allocation • AllocationRun • Metrics

OptimizedAllocator	
Responsibility	**Collaboration**
• Build the model with capacity and demand limits • Solve it and validate the solution • Create allocations from the validated solution • Create the run and ask for its metrics	• Area • ResourceBase • Resource • TravelTimeMatrix • Allocation • AllocationRun • Metrics

AllocationRun	
Responsibility	**Collaboration**
• Record method, time created and solver runtime • Hold its allocations and its one set of metrics • Be stored and fetched by scenario and method	• Allocation • Metrics • Scenario • Comparison

Allocation	
Responsibility	**Collaboration**
• Record base, area and quantity sent • Keep quantity sent within the base's stock • Belong to one run	• ResourceBase • Area • AllocationRun

Metrics	
Responsibility	**Collaboration**
• Calculate average and maximum response time • Calculate coverage, unmet demand, utilisation and travel cost • Use the same calculation for both methods	• Allocation • TravelTimeMatrix • Area • Resource • AllocationRun

Comparison	
Responsibility	**Collaboration**
• Fetch the baseline and optimised runs of a scenario • Work out the difference in each metric • Present the two results side by side	• AllocationRun • Scenario • Metrics